

SYED MUHAMMAD ALI GILANI

Electrical Engineer | PEC Registered Engineer

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PROFESSIONAL SUMMARY

Electrical Engineering graduate specializing in Electronics with a strong academic foundation in telecommunications, communication systems, and embedded electronics. Familiar with GNSS/GPS technologies, wireless communication fundamentals, signal processing, and sensor integration through academic projects. Proficient in MATLAB, Simulink, and technical documentation. Seeking an entry-level Telecommunications, RF, or Network Engineer position.

EDUCATION

Bachelor of Electrical Engineering

Federal Urdu University of Arts, Science and Technology (FUUAST), Islamabad

CGPA: 3.22/4.00

PROFESSIONAL EXPERIENCE

Power Systems Trainee – 32 MW Hydropower Project

- Gained practical exposure to hydropower generation processes and plant operations.
- Developed understanding of turbine, synchronous generator, excitation and governor systems.
- Learned Emergency Shutdown (ESD) procedures and Lockout/Tagout (LOTO) practices

Transformer Testing & Diagnostics Trainee – Hammad Engineering & Transformer Co.

- Developed understanding of transformer oil testing, dielectric strength, moisture analysis and DGA.
- Reviewed transformer condition assessment reports and preventive maintenance practices.
- Assisted in diagnostic procedures used to improve equipment reliability.

CORE COMPETENCIES

- Communication Systems
- Wireless Communication Fundamentals
- GNSS/GPS Technologies
- Networking Fundamentals
- Embedded Hardware
- Signal Processing Fundamentals
- Technical Troubleshooting
- Technical Documentation

TECHNICAL SKILLS

- Engineering Software: MATLAB, Simulink, Proteus, AutoCAD, Arduino IDE, Microsoft Office
 - Communication Systems, Wireless Communication Fundamentals, Networking Fundamentals, GNSS/GPS Technologies, Signal Processing Fundamentals, Embedded Systems, Digital Electronics
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FINAL YEAR PROJECT

GNSS/INS-Based Vehicle Navigation Prototype

- Designed and developed an Arduino Uno-based vehicle navigation prototype integrating GPS and IMU sensors for real-time positioning and navigation.
 - Integrated GPS and sensor modules, implementing embedded navigation logic using Arduino IDE.
 - Performed system calibration, testing, troubleshooting, and technical documentation .
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PROFESSIONAL DEVELOPMENT & LEADERSHIP

- Organizer – YOUTH ROBOTECH
 - Participant – AI INNOFEST (Bahria University)
 - Participant – National Youth Conference (PICSS)
 - Workshops: GNSS/INS Navigation, Line Follower Robot, Home Automation, RC Plane
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LANGUAGES

English | Urdu